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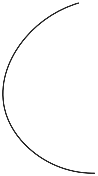

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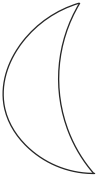

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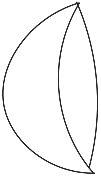

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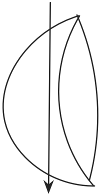

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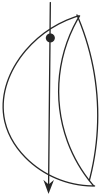

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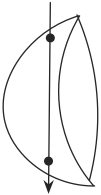

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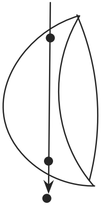

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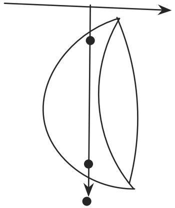

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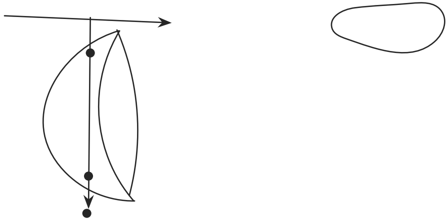

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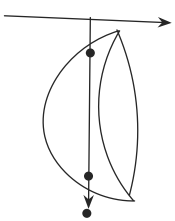

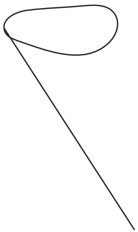

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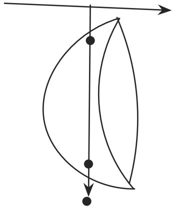

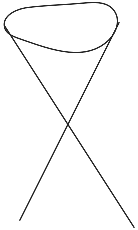

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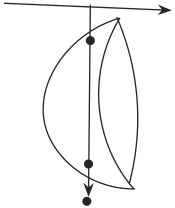

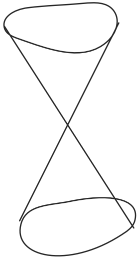

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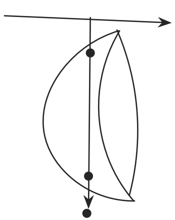

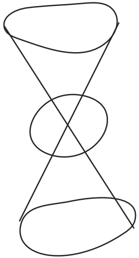

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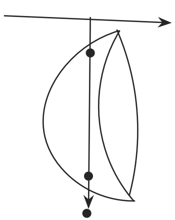

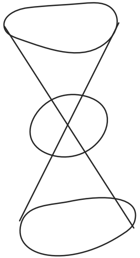

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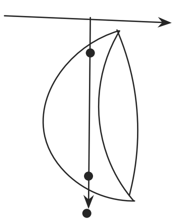

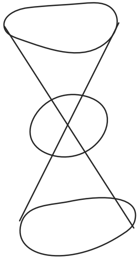

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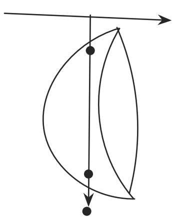

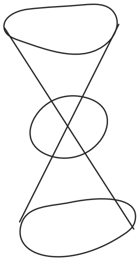

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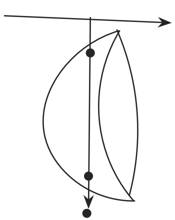

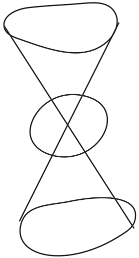

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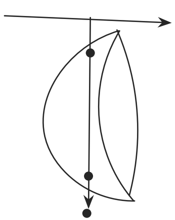

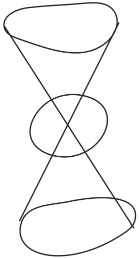

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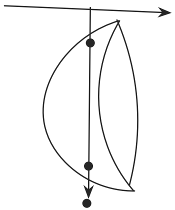

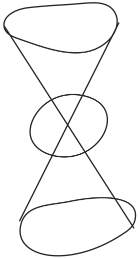

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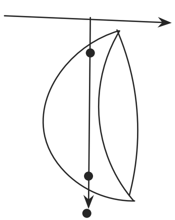

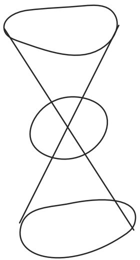

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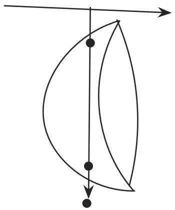

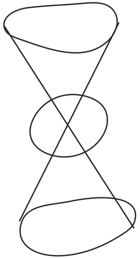

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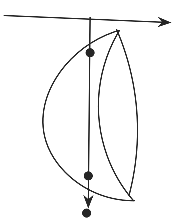

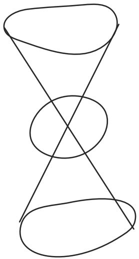

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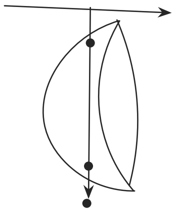

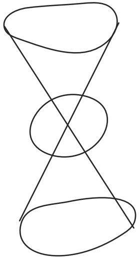

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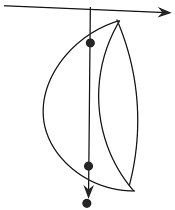

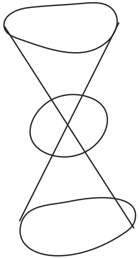

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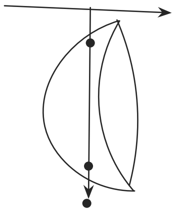

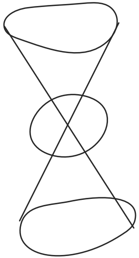

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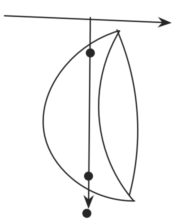

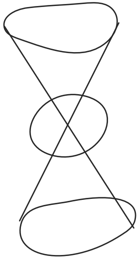

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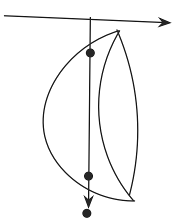

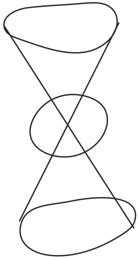

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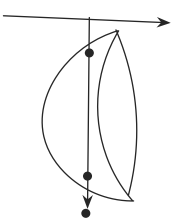

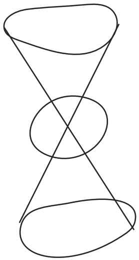

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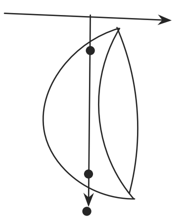

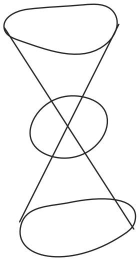

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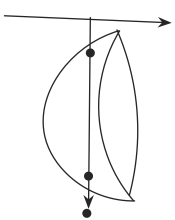

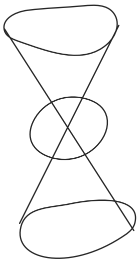

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$$\begin{tikzcd} U_1 \arrow[r, "\sigma_1"] & S \\ & U_2 \arrow[u] \end{tikzcd}$$

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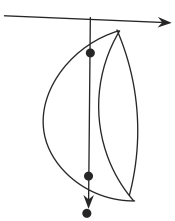

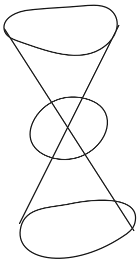

Surfaces

$U \subset \mathbb{R}^2$ is a disk

$\sigma : U \to \mathbb{R}^3$ "smooth" is called a coordinate patch if it is injective

Give an example of a coordinate patch for the sphere

points satisfying $x^2 + y^2 + z^2 = r^2$

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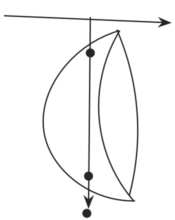

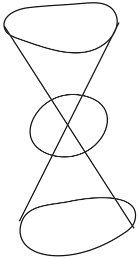

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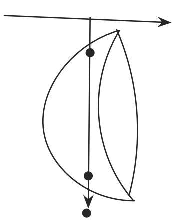

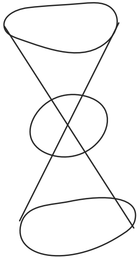

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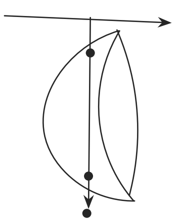

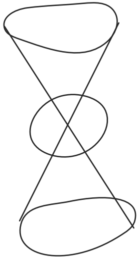

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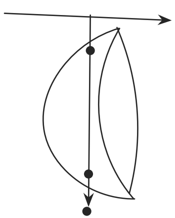

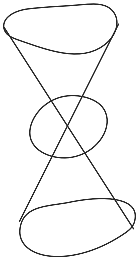

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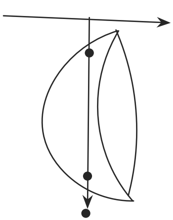

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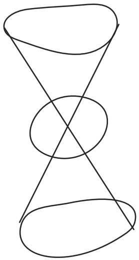

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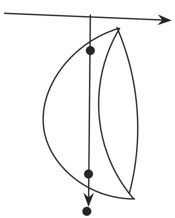

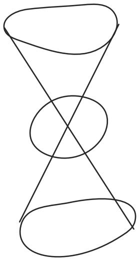

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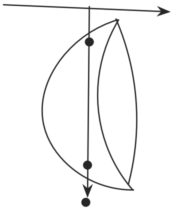

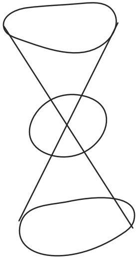

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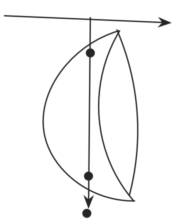

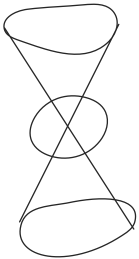

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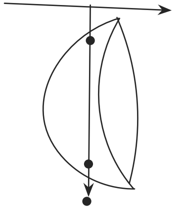

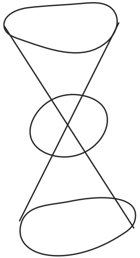

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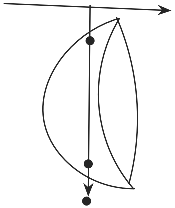

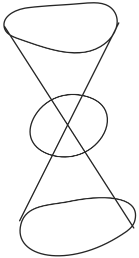

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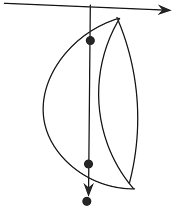

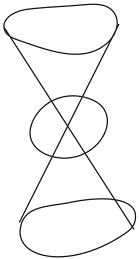

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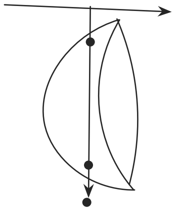

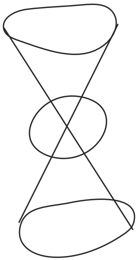

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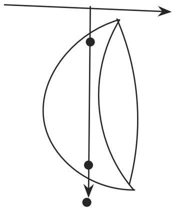

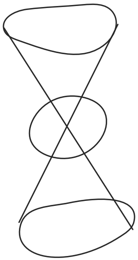

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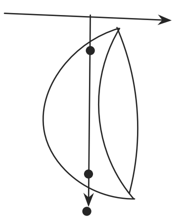

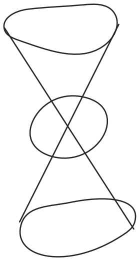

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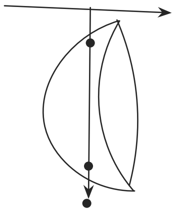

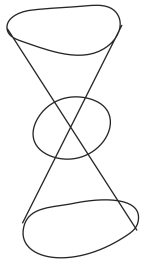

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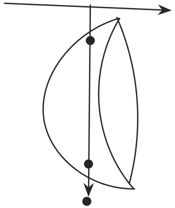

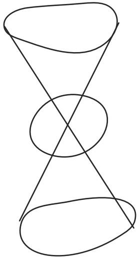

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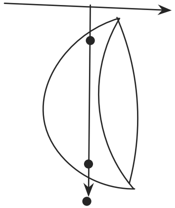

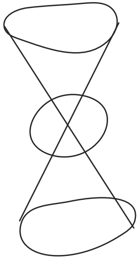

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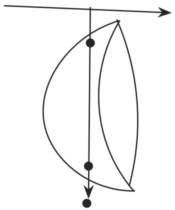

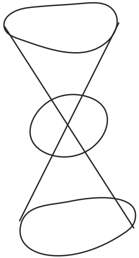

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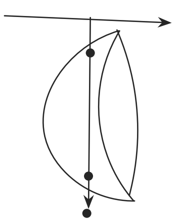

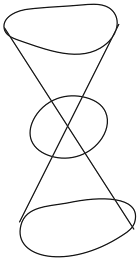

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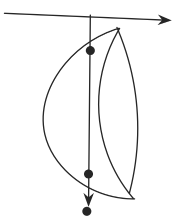

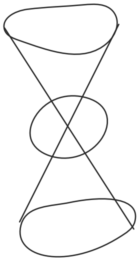

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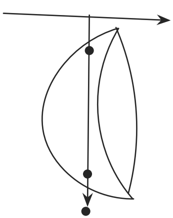

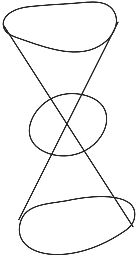

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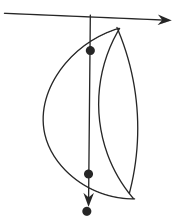

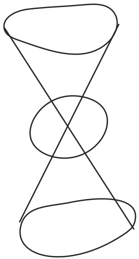

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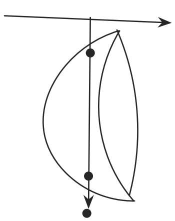

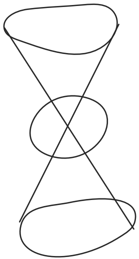

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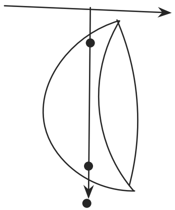

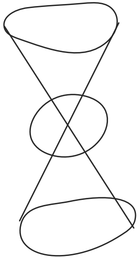

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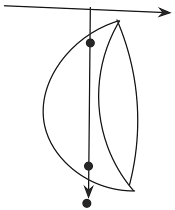

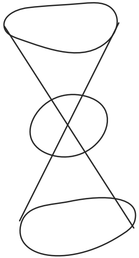

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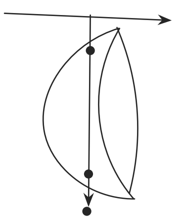

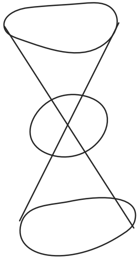

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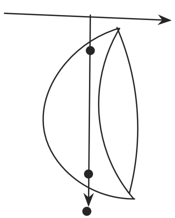

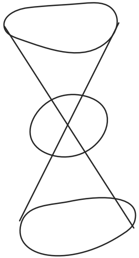

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$g'(x) = \lim_{h\to 0} \frac{g(x+h)-g(x)}{h}$

Suggestion: define the deritivative of f in the same way
$\lim_{(h_1,h_2)\to(0,0)} \frac{f(x+h_1,y+h_2)-f(x,y)}{h_1+h_2}$
Or, we could
$\lim_{(h_1,h_2)\to(0,0)} \frac{f(x+h_1,y+h_2)-f(x,y)}{\sqrt{h_1^2+h_2^2}}$ so that we are dividing by the length
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$= 2y$
Similarly, If $h_2 = 0$ always and $h_1 \to 0$ we will get $2x$

How about, finding the rate of change along a curve
Consider a curve parametrized by $\theta : (a,b) \to \mathbb{R}^2$
$\theta(t) = (x(t), y(t))$
We will define the derivative of f along θ to be the derivative of $f(\theta(t))$ i.e. $f(x(t), y(t))$
Note that $f \circ \theta : (a,b) \to \mathbb{R}$

Consider the point (α, β) and the curve $\theta(t) = (t, \beta)$
What is the derivative of f along *this theta*? i.e. The derivative of $f \circ \theta$ at $t = \alpha$ is
The partial derivative of f at (α, β) is defined as (denoted $\frac{\partial f}{\partial x}$)
$\lim_{t\to\alpha} \frac{f(t,\beta)-f(\alpha,\beta)}{|t-\alpha|}$
Similarly, we can define the partial derivative of f with respect to y, denoted $\frac{\partial f}{\partial y}$ by using the curve
$\theta(t) = (\alpha, t)$

Multivariable Chain Rule
$\frac{\mathrm{d}}{\mathrm{d}t} f(\theta(t)) = x'(t)\frac{\partial f}{\partial x} + y'(t)\frac{\partial f}{\partial y}$

Digression (Multivariable calculus)
$f : \mathbb{R}^2 \to \mathbb{R}$
$f(x,y) = x^2 + y^2$
How would you define the "derivative" of f?

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If two curves pass throught the same point and they have the same velocity at that point, then the derivative of f along either curve at that point will be the same

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